مدارس الإمارات الوطنية
Emirates National Schools

Grade: 4 | Term 1, Week 10 (2 Nov to 6 Nov) | TCD: Future Skills 4: Design, Tools and Simple Solutions | Session length: 1 hr 30 min

T1 WEEK: 10	**Lesson 1: Matatalab Pro Refresh and Angle Blocks**
Standard(s):	Math: 4.MD.C.5 (angles as turns, measuring angles in degrees); Technology: algorithms and sequencing
Literacy Strategies:	Word wall with sequence, algorithm, angle, degree and debug; partners use the frame: First the robot ..., then it turns ... degrees.
Assessment:	Observation of coding with a checklist; shape challenge results; live quiz score on the TCD website.
WALT (Learning Target):	We are learning to program precise turns using angle blocks.
WILF (Success Criteria):	I can build a sequence of movement blocks and run it. I can explain that a full turn is 360 degrees and a quarter turn is 90 degrees. I can use an angle block to make the robot turn exactly the amount I need. I can fix my code when the robot does not do what I planned.
Targeted Vocabulary	algorithm, sequence, angle, degree, quarter turn, half turn, command tower, debug
Resources	Matatalab Pro Sets (one per pair or group of three), Command Towers and MatataBots, charged, Map mats, Artist add-on with pens, A3 paper for drawing, Angle cards (45, 90, 120 degrees), Protractors Accompanying resource: Grade4_Week10_Presentation.pptx (slides, hook video and live quiz code on the TCD website), Grade4_Week10_Worksheet and Grade4_Week10_Lesson_Plan.
National Identity Integration	Link angles to the geometry in Emirati architecture: the patterns on the Al Fahidi wind towers and mosque windows are built from precise angles.
AI/Digital Learning	Matatalab Pro Sets; hook video; live quiz on the TCD website.
Direct Instruction: I Do	Refresh the coding blocks and the Command Tower (slide 6). Teach degrees with a protractor and body turns (slide 7). Model using an angle block to draw a square on A3 paper (slides 8 and 9).
Guided Practice: We Do	Pairs run a short refresh route on the map mat and fix one bug together. Class plans the triangle code together on the board: how many turns, and how big?
Independent Practice: You Do	Pairs complete the shape challenge and record their code on the worksheet (slide 11). Challenge levels on the board (slide 12). Live quiz on tablets (slide 13).
Differentiated Instruction: Beginning / On / Above	**Beginning: Draw the square only, using a ready-made code card to copy from.** **On: Draw the square and the triangle, planning the code yourself.** **Above: Draw a hexagon (60 degree turns) or a star and explain the turn you used.**
Closure/Reflection	Two pairs show their drawings and explain their turn angles. Exit ticket: one thing an angle block does (slide 14). Pack away sets and check pens are capped.
Inclusion / EAL Support	☑ Visual Supports ☐ Chunked Instructions ☐ Reduced Language Load ☑ Model / Worked Example ☐ Pre-teach Vocabulary ☐ TA/LSA Support ☐ Other: ☐ Extra Processing Time ☐ Text Read by TA/LSA ☑ Flexible Grouping ☐ Assistive Technology ☐ Movement / Sensory Break ☑ Scaffold / Sentence Starters

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G&T / Extension	☑ Extension / Challenge Task ☐ Real-World Application ☑ Higher-Order Questioning ☐ Curriculum Compacting ☐ Depth & Complexity Task ☐ Above-Grade-Level Texts ☐ Independent Inquiry ☐ Student-Led /Passion project ☐ Leadership / Peer Coaching ☑ Open-Ended Task ☐ Other: _____